

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and built reusable components for a medical device's firmware, translating engineering specifications into functional code, which reduced integration time by **15%**.
- Developed and optimized data processing pipelines using **Python** and **SQL** to support analytical reporting for medical device performance, cutting data retrieval times by **30%**.
- Authored detailed technical documentation, including **work instructions** and **Standard Operating Procedures (SOPs)**, for new product development processes, ensuring clarity and adherence across 3 engineering teams.
- Executed rigorous lab bench testing and simulated use validation activities for medical device components, identifying and resolving **5 critical design flaws** prior to pre-clinical studies.
- Assisted in the validation of new engineering test methods, preparing protocols and reports that supported compliance with **ISO 13485** standards and internal quality guidelines.
- Contributed to the development and qualification of new process equipment, performing hands-on testing and supporting vendor selection for a key manufacturing component, resulting in a **10% increase** in throughput.

SKILLS

Medical Device Development – Design, Process Development, Validation Protocols, Test Method Development, IQ/OQ/PQ, Regulatory Compliance, Design Inputs, Clinical Use Scenarios

Programming & Scripting – Python, Java, SQL, Shell Scripting

Data Analysis & BI – Power BI, Tableau, DAX, Power Query, R, Statistical Analysis, DOE, Data Warehousing

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Enterprise Systems – ERP Support, REST APIs, PostgreSQL, MySQL, Oracle

Methodologies & Documentation – Agile/Scrum, SOPs, Technical Writing, cGMP, GLP, Six Sigma Concepts

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

**BACHELOR OF ENGINEERING in Computer Science – SNS College
of Technology, Coimbatore, India**

01/2021

PROJECTS

Automated Medical Device Component Validation Framework

- Developed an automated testing framework using **Python** and integrated it with existing lab equipment to streamline the validation of critical medical device components, completing testing cycles **40% faster**.
- Implemented data logging and analysis capabilities using **Pandas** and **SQL**, generating automated reports that summarized test results, identified failure modes, and provided actionable insights for design iteration.

- Designed and documented new test procedures and work instructions as part of the framework development, ensuring compliance with **FDA QSR** and **ISO 13485** quality standards.

Financial Services Transaction Monitoring Dashboard (HCL Technologies)

- Enhanced a payments operations dashboard by implementing real-time transaction monitoring and alerting for failed payment messages, improving operational awareness and reducing incident response time by **20%**.
- Developed automated Python scripts for data validation and reconciliation workflows, ensuring data integrity across multiple financial system integrations and supporting audit trail requirements.
- Collaborated with cross-functional operations and business analyst teams to gather requirements and provide post-implementation support for dashboard features, ensuring alignment with client needs and regulatory awareness.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)